UPCI Summer Program Evaluation Report

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University of Pittsburgh

EXECUTIVE SUMMARY

The University of Pittsburgh Cancer Institute (UPCI) Academy strives to provide cutting edge research and career preparatory experiences to a diverse group of highly motivated high school students who are pursuing higher education and careers in STEM fields, especially research and medicine. The present evaluation report aims to describe the student population that participated in the summer program in order to provide key insights to improve the program's future iterations. Furthermore, we also provide information of how this program affected students' career goals specifically towards research and medicine.

In total, 59 students (32 women and 27 men) were surveyed at three different time points:

1) On entry, students completed a Pre-Reflection survey aimed to understand their career goals, motivation for attending the program, and demographic information
2) Students also completed a motivation survey during their first week to assess their science identity levels.
3) Finally, students completed an exit survey to reflect on how their career stances, research skills, and understanding of how science works changed during the program.

The KEY findings regarding **ENTERING CHARACTERISTICS** are:

- 38% of participants are intending to be PRE-MED students
- There are not important differences between students with and without prior research experiences in their motivation to participate in UPCI. The biggest difference is that students with prior research experiences are more focused on deepening their knowledge of STEM topics, while students who are participating in research for the first time are more focused on understanding how science works.
- Not surprisingly, on average, students that have had at least one prior research experience feel more confidence in their research skills across every skill that was examined.
- Students allocated to the Computing Research Experiences are less likely to have stipends (i.e., tend to have a higher socio-economic background), always have prior coding experiences and report uniformly feeling comfortable with coding
- Students that participated in the UPCI experience have on average high Science identities, especially in Biology identities. However, students participating in Computing Research Experiences were similarly high on biology, chemistry, physics, and math identities.

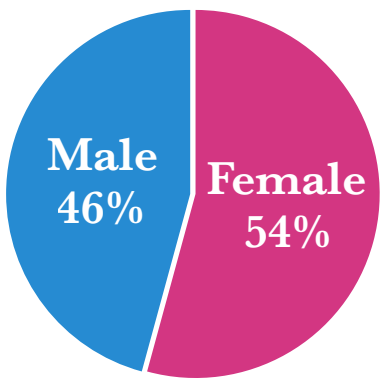
The KEY findings regarding students' EXIT CHARACTERISTICS are:

- Nearly all participants report that their idea of a "a career scientist" become more clear after your research experience, including female students and those students without prior research experiences.
- The program positively influenced students' future plans to attend Medical School or Graduate School in STEM. Moreover it impacted students that have not considered Graduate School before and shifted them towards STEM and health careers. These patterns are similar for female students and students without prior research experiences.
- Students reported an improvement in their research skills.
- Students reported positive support from their research mentors and felt they were meaningful contributors to their research team.

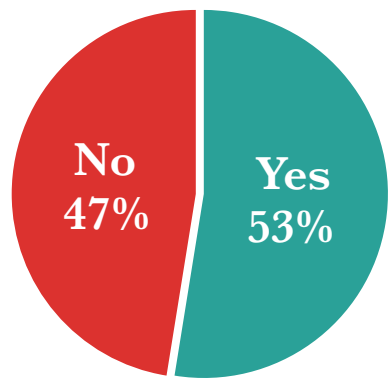
Conclusions

In conclusion, the UPCI academy helped develop participants' research skills and influenced their future career goals. In general, the program was successful at meeting its mission goals. However, there were a few isolated cases with low ratings, and we recommend contacting these students to gather further information as to why their experience was not satisfactory. It is possible that the problem lies within particular mentors and is a matter of improving mentoring resources for students and participant faculty.

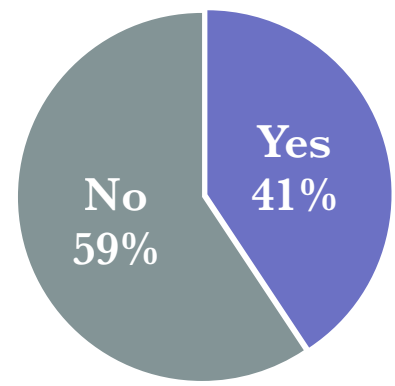
STUDENTS' ENTRY CHARACTERISTICS



Gender



First Time Research Experience



Stipend Recipient

Why are they here and how their motivations differ by Prior Research Experience?

	First Time Research Experience	Prior Research Experience
High in both groups		
I am considering a degree or career in computer science, biology, or biomedical science	42%	37%
I am interested in and what like to learn more about computer science, biology, or biomedical science	42%	37%
I am considering a degree or career in engineering	19%	14%
Higher in First Timers		
To see if I enjoy doing research	46%	20%
To understand how scientists do science	41%	25%
This will look good on college applications	27%	20%
Low in both groups		
I am considering a degree or career in another STEM field	8%	7%
This will look good on medical school applications	7%	10%
This will look good on graduate school applications	5%	5%

The motivations of students don't differ much by whether they had Research Experience before or not except that students that have never done research before are more likely to do this experience to understand how to do science and whether they enjoy research. Of the students that have had at least one prior research experience, career goals are the main drivers to want to participate in UPCI.

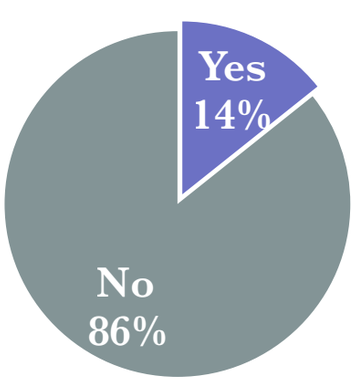
How confident are they on their Research Skills by Prior Research Experience?

Percentage of students that feel confident about their abilities (rating above a 5) to… (sorted from most to least confident)

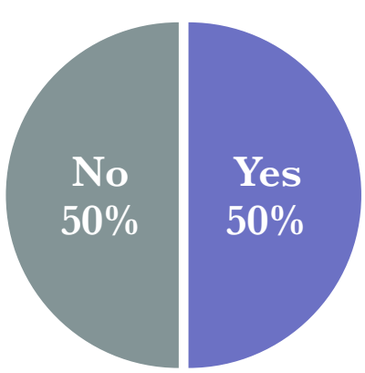
	First Time Research Experience	Prior Research Experience
Critically analyzing the quality of data generated in my own research	45%	86%
Communicating scientific results to non-scientists	58%	82%
Gathering and organizing data through the use of scientific tools and technology.	48%	82%
Making use of primary research literature (e.g. journal articles) to understand current advances in a scientific field	48%	75%
Communicating a critical thesis that clearly establishes the focus of my position on an issue	45%	75%
Understanding how to pursue careers in science.	45%	75%
Communicating with scientists using a scientific vocabulary	35%	64%
Using software to create scientific models to make predictions	16%	43%

Not surprisingly, on average, students that have had at least one prior research experience feel more confidence on their abilities on all the above metrics. Therefore, one of the future goals of the UPCI program should be to specifically focus on students that have never had prior research experiences. However, it is important to note that neither group of students feel confident on their skills to create scientific models to make predictions using software. Given this is a really specific skill we looked at how students specifically placed at the CoSBBI and DisCoBio sites rated their ability on this specific metric

Confident on using software to create scientific models to make predictions (CoSBBI and DisCoBio students only):



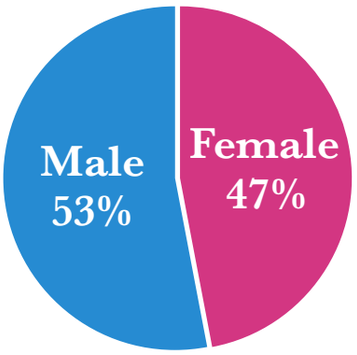
Given that students placed at these sites seem to be slightly different from the other we will look at this subset of students and analyze their entry level characteristics.



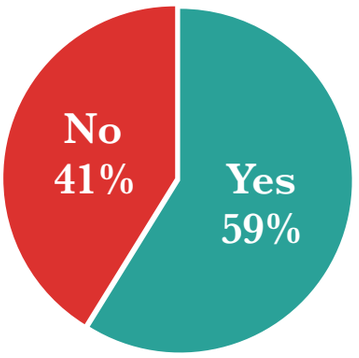
First Time Research Experience

Prior Research Experience

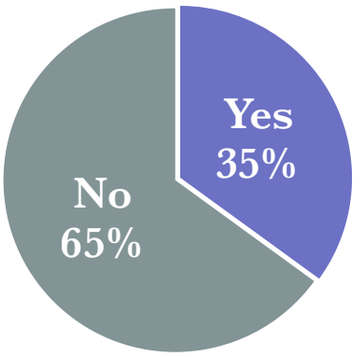
Who is most likely to be placed in a Computing Research Experience?



Gender



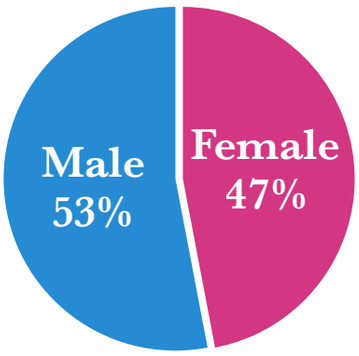
First Time Research Experience



Stipend Recipient

Looking at the overall sample of students that participated in UPCI, the distribution on each of these categories was 50/50. However, when looking at the subset that participates on research where programming is required, we see that students with a stipend are less likely to select this research experience. Furthermore, students that have had research experiences before were also more likely to chose it.

Of those who do…



Proficient in at least one coding language

53% of the allocated students report that they are proficient in at least one coding language. There is not gender difference as to whether students identify themselves as proficient in at least one coding language.

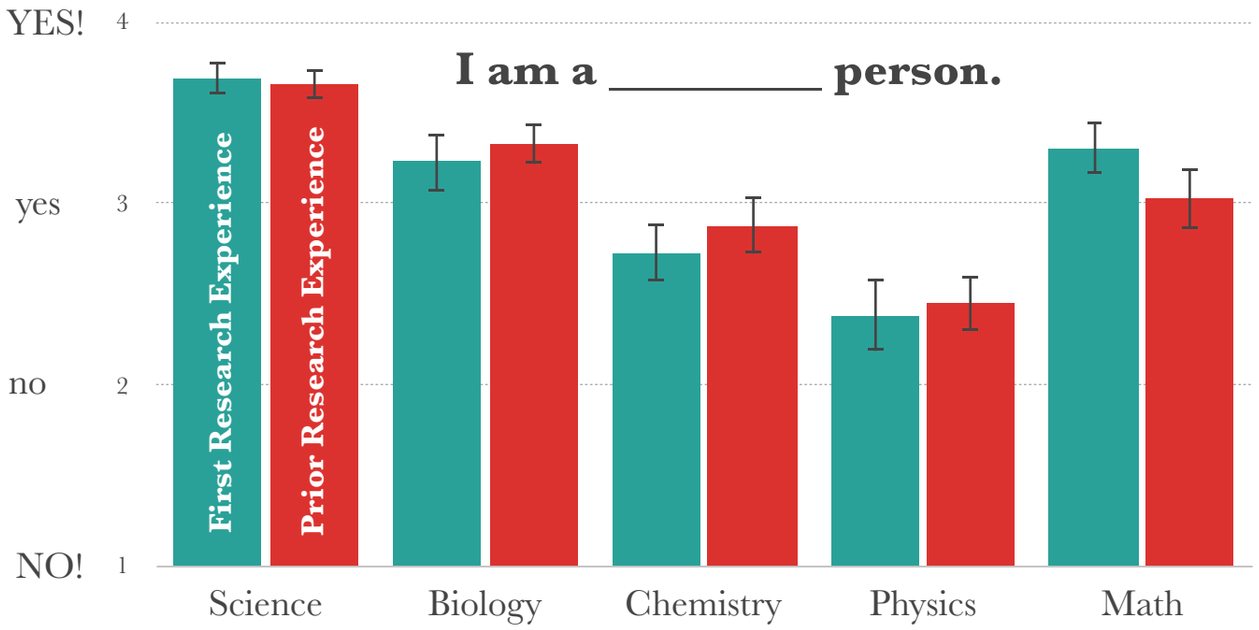
82% Report that they learned coding at school.

23% Have taken coding classes outside of school.

6% Report they are proficient in multiple coding languages.

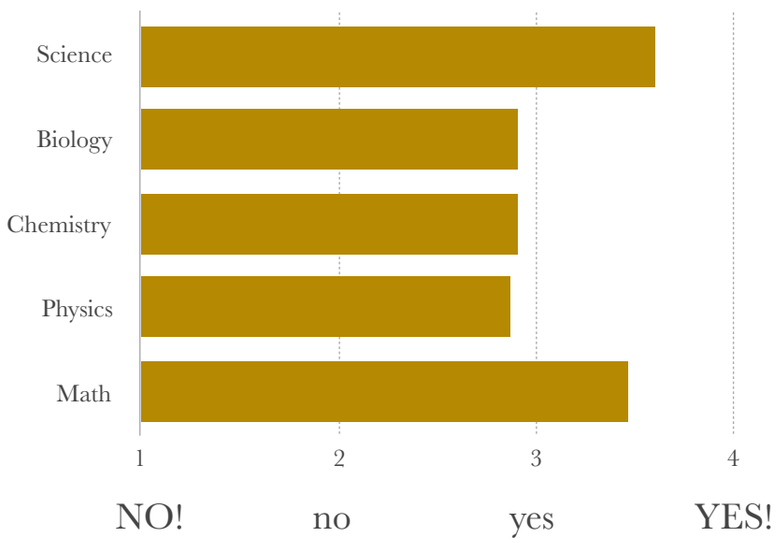
82% Report they want to major in biology.

Students' Science Identity by Research Experience



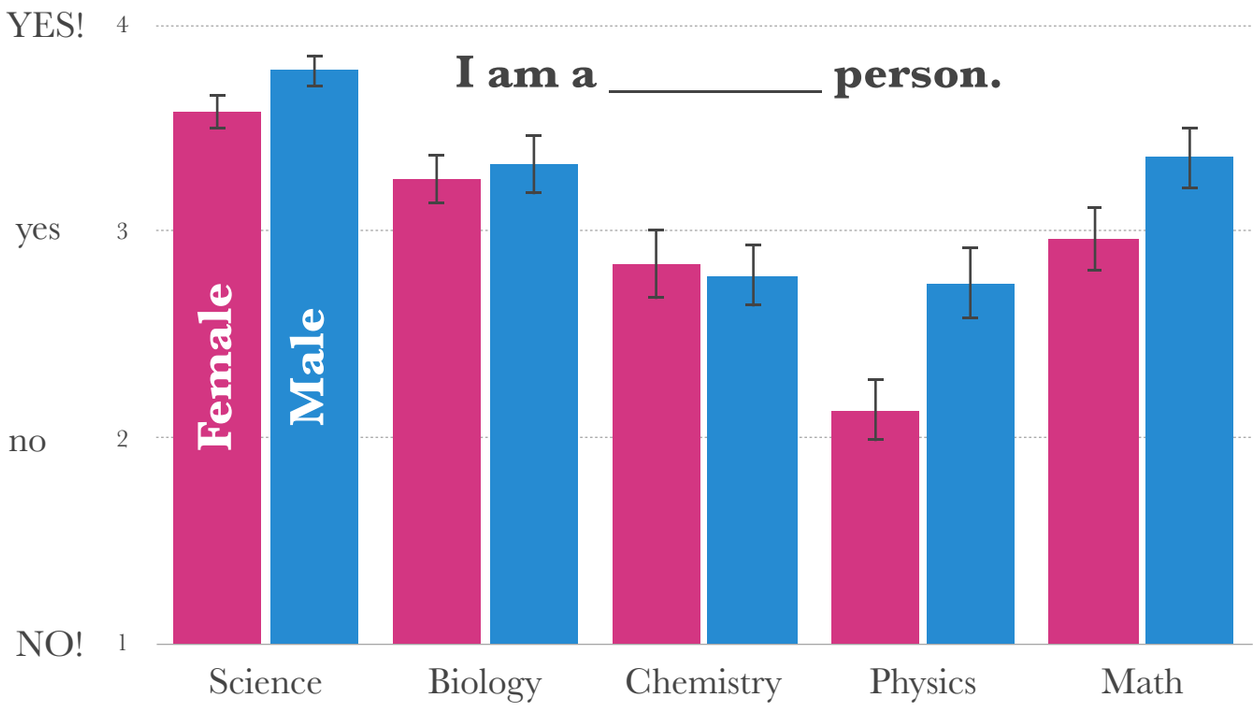
There is not much difference between students that had prior research experiences and those who don't on whether they identify as science people or not . This doesn't mean prior experiences have not shifted these identities but that the particular scale has a ceiling effect. Of things to note is that overall these students have high science identity, but when we look at particular topical identities the endorsement of these identities is very different. Students that participated at UPCI have:

1) A **VERY HIGH** endorsement of a Biology identity
2) **High** Chemistry identity
3) They on average have a **NEUTRAL** stance on a Physics identity
4) The only statistically significant difference between students that had Prior Research Experiences and those who never had Research Experiences before in on Math Identity. Students with Prior Research Experiences have slightly lower math identities, though still on the **HIGH** range.



Is important to note that for the students placed on Computing Research Experiences, physics identity is higher on average. That is, the topical identities on average are more balanced for this students than on the overall population.

Students' Science Identity by Gender



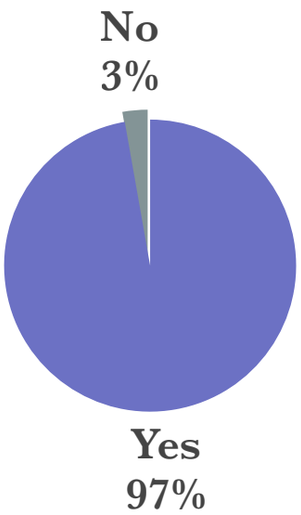
There are more important differences on identities by gender than by research experience:

1) Male students have **HIGHER** science identity on average than do female students.
2) There are no statistically significant differences for biology and chemistry identities.
3) Men have **HIGHER** physics identity. Moreover, men on average have a **HIGH** physics identity and women on average have a **LOW** physics identity,
4) Though both group of students have **HIGH** math identity, men identify as math people more than do women.

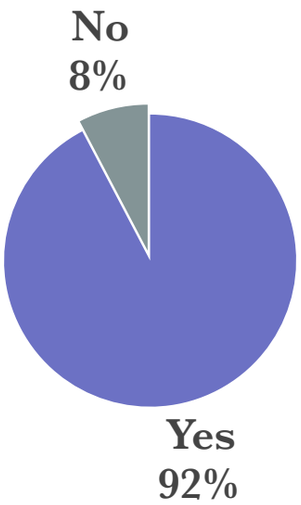
It is important to note that students that participated at UPCI have high identities on average; that is, the gender differences except on Physics are not big enough to cause concern. Furthermore, the UPCI experience is more likely to target students' biology and chemistry identities.

STUDENTS' EXIT CHARACTERISTICS

Has your idea of a "a career scientist" become more clear after your research experience?

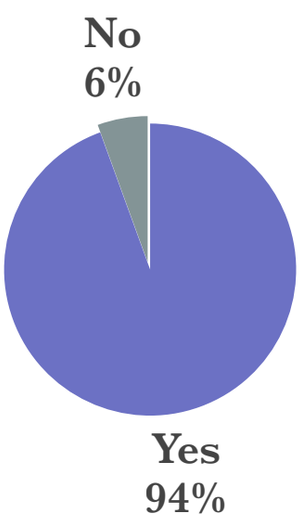


First Time Research Experience

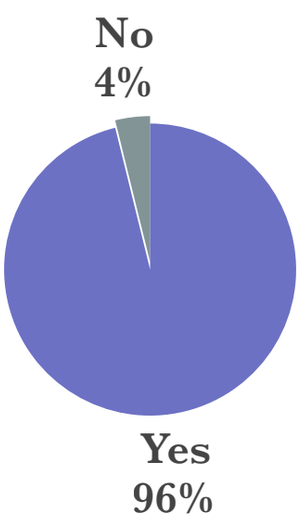


Prior Research Experience

Nearly all students, no matter their prior research experience, reported that their idea of a "career scientists" became more clear after their experience at UPCI.



Female



Male

Moreover, this does not differ by gender, both men and women report that their idea of a "career scientists" became more clear after their experience at UPCI. This gender results don't differ by whether students have had prior research experiences or not.

How have participant's plans Post-College changed after UPCI?

	Before	After
Have not Considered GradSchool	16%	3%
Medical Degree	38%	43%
GradSchool in STEM	35%	43%
GradSchool in Non-STEM	3%	6%
Other	8%	2%
Planning not to pursue GradSchool	0%	3%

The results show that UPCI has been really successful at influencing student's plans to applying to GradSchool on STEM fields and pursuing a Medical Degree. The percentage of students that had not considered GradSchool before this experience decreased dramatically as well as those who had Other plans. We will now look at how the program influenced to particular groups: First-Research Experience students and women.

First-Time Research Experience Participants

	Before	After
Have not Considered GradSchool	22%	6%
Medical Degree	36%	45%
GradSchool in STEM	28%	37%
GradSchool in Non-STEM	5%	8%
Other	9%	1%
Planning not to pursue GradSchool	0%	3%

The pattern is very similar for students that have not had Research Experiences before. However, this does not make the results less important. Given UPCI is the first time these students have had the opportunity to participate in research is pretty impressive that it created such an impression that the number of people that have never considered grad school before decreased so dramatically.

Female Participants

	Before	After
Have not Considered GradSchool	11%	0%
Medical Degree	36%	51%
GradSchool in STEM	39%	43%
GradSchool in Non-STEM	3%	5%
Other	11%	0%
Planning not to pursue GradSchool	0%	0%

The pattern for female participants is slightly different. First of all, they are less likely to not have considered GradSchool before participating at UPCI. Moreover, the program shifted all of the female participants that had not considered GradSchool or considered Other opportunities towards STEM or Medical Degree goals.

How have participant's perceptions of their Research Skills changed after participating at UPCI?

Percentage of students that feel confident about their abilities (rating above a 5) before and after the academy… (sorted from most to least confident)

	Before	Now
Communicating with scientists using a scientific vocabulary	23%	87%
Making use of primary research literature (e.g. journal articles) to understand current advances in a scientific field	25%	83%
Gathering and organizing data through the use of scientific tools and technology.	48%	82%
Critically analyzing the quality of data generated in my own research	28%	82%
Communicating scientific results to non-scientists	37%	82%
Understanding how to pursue careers in science.	28%	82%
Communicating a critical thesis that clearly establishes the focus of my position on an issue	33%	78%
Using software to create scientific models to make predictions	17%	53%

There are a couple of interesting things from this result:

1) The Before percentages upon reflection are lower than those on Pre-Reflection. This only show that students overrated their abilities before coming into the academy and after participation they were able to make a better assessment of where they were and how much they grew.
2) Overall this is a very positive result, not only students feel more confident now they didn't feel discouraged through this learning process (as they are now more likely to want to go to GradSchool in STEM or Medical School).
3) Finally, the increase in confidence using create scientific models to make predictions increased similarly for students on all sites (see table below for students specifically placed at the CoSBBI and DisCoBio).

	Before	Now
Using software to create scientific models to make predictions	16%	61%

How do participants rate their experience at UPCI?

The participants rated their experiences on their specific labs from Not True to True on a 5 point scale. The scale was scored from -2 to 2. A mean close to zero means a neutral stance towards the statement. We proceeded to compare the means of the statements by the following groups: Gender and Research Experience.

First-Time Research Experience Participants

	First Time Research Experience	Prior Research Experience	p-value
My responsibilities and assignments were clearly defined	0.84	1.08	0.70
I was given constructive criticism in a professional manner	1.21	1.11	0.35
Average of I feel that I made a positive contribution to my research group	0.81	0.96	0.66
My research experience increased my confidence in my ability to be a contributing member of a research team	1.03	0.92	0.53

There are no statistically significant differences as to how, on average, students reported their experience. This is important because students participating in a research experience for the first time are more likely to feel they could not be good contributors on a research team. All ratings were also around a value of 1 meaning that in general students are feeling supported and given enough guidance in their research projects.

Female Participants

	Female	Male	p-value
My responsibilities and assignments were clearly defined	1.05	0.84	0.37
I was given constructive criticism in a professional manner	1.08	1.32	0.34
Average of I feel that I made a positive contribution to my research group	0.88	0.92	0.88
My research experience increased my confidence in my ability to be a contributing member of a research team	0.91	1.12	0.39

There are no statistically significant differences as to how, on average, students reported their experience by gender. However, some students did make negative ratings. It may be necessary to look at these particular cases and find out why they didn't feel successful and supported.